

TABLE I
SOME NETWORKS WITH STANDARDS OF PACKETS/FRAMES

Network	Specification	Packet/frame size
Ethernet/LAN for office automation/IEEE 802.3	IEEE (Institute of Electrical & Electronics Engineers)	• Minimum: 64 bytes • Maximum: 1522 bytes of frame • Data/payload/packet size: 46 to 1500 bytes
Manufacturing Automation Protocol/IEEE 802.4	IEEE	• No minimum frame size • Maximum frame size: 8191 bytes
X.25	ITU-T (International Telecommunication Union)	• 128 to 256 bytes of packet
Synchronous data link control/High level data link control (SDLC/HDLC)	International Business Machines (IBM)	• Data/Payload/Packet: 46 to 1500 bytes • Overhead: 6 to 9 bytes for SDLC, and up to 20 bytes for HDLC

delay, but voice cannot. This makes any data, while on transmission from source to destination, to be possibly chopped into pieces (each piece called a packet) and stored at intermediate nodes, to be transferred independently and subsequently, unlike a stream transfer of voice over a telephonic system.

To meet the characteristics of data, a branch of engineering has been developed for data transmission, giving rise to a specialised area called data networks. Data networks carry data of any base, such as decimal, octal, hex and so on.

Computer networks are data networks for carrying binary data made of 0s and 1s. Although technological and engineering aspects are almost the same, computer networks are an element of a set of data networks. The field of data networking began with the concept of data by packets. This gave rise to different switching techniques for data transport.

Switching techniques

Fig. 1 portrays the different switching techniques available for various transport technologies. Circuit switching fundamentally works with establishing a prior and dedicated link between source and destination for full duration of communication. After and when communication between parties is over, the said link is delinked for possible use of other potential users.

Circuit switching is appropriately used in telephony networks. In store and forward switching, messages are stored at intermediate nodes available between source and destination, and subsequently forwarded till the desti-

nation is reached. Pre-establishment of a dedicated link between source and destination is not necessary. Links are not dedicated and can be shared among all potential users.

Store and forward switching are of two types: message switching and packet switching. In message switching, the message is taken as one unit for transport between nodes. Message switching is appropriately applied to telegraphy and e-mail communication.

Message switching has a few limitations, as follows:

- When a message is long, junking of the link creates problems in communication.
- Due to limitation of buffer at intermediate nodes, a long message may result in either loss of data or buffer congestion.
- In a long message, if the link is junked, priority assigned transmission may not be possible.

These limitations are duly-solved with packet switching. In packet switching, messages or data are fragmented or divided into many parts, each called a packet. Each packet is enveloped with some overheads for packet numbering, error control, source address, destination address, start of frame, end of frame, priority assignment, synchronisation and so on, to make a frame. Packets/frames are made to traverse in the network from node to node, independently, to reach the destination from source via best possible/optimised shared link(s).

Packet switching is inherently fast switching. Due to packetisation (availability of packets that may move independently) and shared links between source and destination (availability of parallel links), different packets may traverse in different parallel paths. This results in lower transmission time.

However, overhead bits of frames/packets require more transmission time over original data. Thus, tradeoffs exist between parallel transport with overhead bits. Fewer number of packets amounts to possibility of use of fewer number of more parallel links, causing slower transport. More numbers of packets mean use of more parallel paths, resulting in faster transport. And more packets use more overhead bits, causing more transmission time, lesser of fast transport.

How many packets should be used and what should be the optimal size of each packet to derive maximum benefit out of packetisation? No specific engi-

TABLE II
COMPARISON BETWEEN LAN, MAN AND WAN

Network	Coverage area	Bandwidth	Links	Cost	Specialities
LAN	Typically, 1km; over a building, an institution, etc	Low	Ethernet cable	Low	• Fully-private network • Shared media network • Can support 100% resource sharing
MAN	Typically, 100km; over a city, zonal district, etc	Medium	Coaxial cable, microwave link	Medium	• Zonal public network • Switched network
WAN	Typically, over 100km to 10,000km; over a country or province	Highest	Satellite links, telephonic links	Most expensive	• National public network • Switched network
Internet	Beyond 10,000km; over multiple countries, intercontinental (planets)	Highest	Logical connectivity using physical networks	Not so expensive	• Logical connection across the globe